

PneumoVision: Pneumonia Detection from Chest X-Rays Using Classical Machine Learning

Phase III: Final Design, Implementation, and Results

Ahmed Ahmed Mokhtar

Ahmed Mohamed Abdelsalam

Alhussien Ayman

Abstract—Pneumonia is a severe respiratory infection that requires rapid and accurate diagnosis, typically conducted via chest X-rays. Interpreting these X-rays can be prone to human error and relies heavily on the availability of expert radiologists. The objective of this project is to build an automated, interpretable classical machine learning pipeline capable of classifying chest X-rays into 'Normal' or 'Pneumonia' categories. During development, two distinct pipelines were evaluated: a Traditional ML Approach and a refined Clinical Imaging Approach. In a clinical diagnostic setting, minimizing False Negatives is the absolute highest priority. Therefore, the primary evaluation metric for this system is Sensitivity (Recall for the Pneumonia class), ensuring that patients with the disease are correctly identified and receive necessary treatment.

Index Terms—Pneumonia Detection, Chest X-Ray, Machine Learning, CLAHE, HOG, GLCM, Computer Vision

I. DATASET DESCRIPTION

The system utilizes a publicly available Kaggle dataset of Chest X-Ray Images. The data was organized into three splits:

- **Training Set:** 5,216 images (1,341 Normal, 3,875 Pneumonia)
- **Testing Set:** 624 images (234 Normal, 390 Pneumonia)
- **Validation Set:** 16 images (8 Normal, 8 Pneumonia)

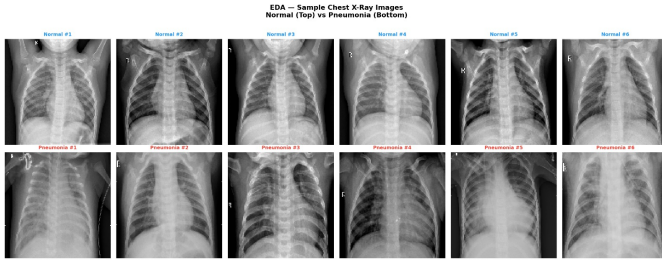


Fig. 1. Directory and hierarchical structure of the Chest X-Ray dataset.

II. METHODOLOGICAL EVOLUTION: APPROACH COMPARISON

To achieve the optimal diagnostic tool, the project iteratively developed and compared two distinct processing pipelines.

A. Approach 1: Traditional ML Pipeline

The initial approach treated the X-rays as standard classification images. Images were resized to a low-resolution

128×128 footprint without targeted noise handling. Feature extraction involved a combination of HOG, Local Binary Patterns (LBP) for local textures, and basic statistical features (skewness, kurtosis). To handle the high dimensionality, Principal Component Analysis (PCA) was applied to retain 95% variance, and Synthetic Minority Over-sampling Technique (SMOTE) was used to balance the dataset.

B. Approach 2: Clinical Imaging Pipeline (Final)

The refined approach prioritizes medical relevance. Images are kept at a higher 256×256 resolution. Instead of analyzing the whole image, CLAHE and Otsu thresholding isolate the lung fields specifically. LBP was replaced by GLCM to capture spatial tissue structure, and PCA was discarded entirely to prevent the loss of critical variance. Instead of generating synthetic noise via SMOTE, class imbalance was handled via native algorithmic weighting (e.g., Balanced Random Forest).

III. FINAL SYSTEM DESIGN & ARCHITECTURE

Based on the iterative evaluation, Approach 2 was selected for the final architecture.

A. Image Preprocessing

To isolate the lung cavities and remove background noise, the pipeline applies Contrast Limited Adaptive Histogram Equalization (CLAHE), followed by Gaussian blurring and Otsu's thresholding. Morphological operations generate a clean binary mask to isolate the lung fields.

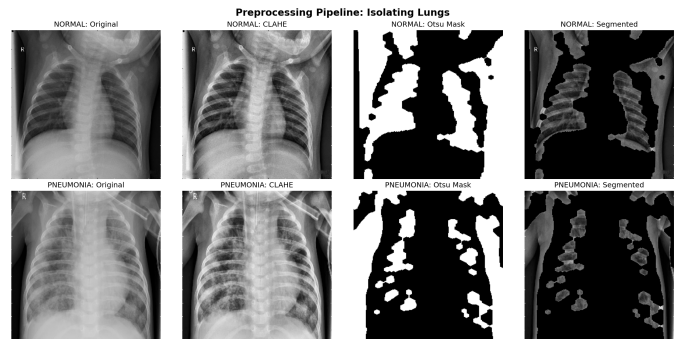


Fig. 2. The image preprocessing pipeline demonstrating CLAHE enhancement and lung masking.

B. Feature Extraction

We extracted an 8,140-dimensional feature vector per image by concatenating two descriptors:

- **HOG:** Used to map structural edges and shapes within the lung cavity.
- **GLCM:** Used to capture textural features and cloudiness (opacities) typical of pneumonia infections.

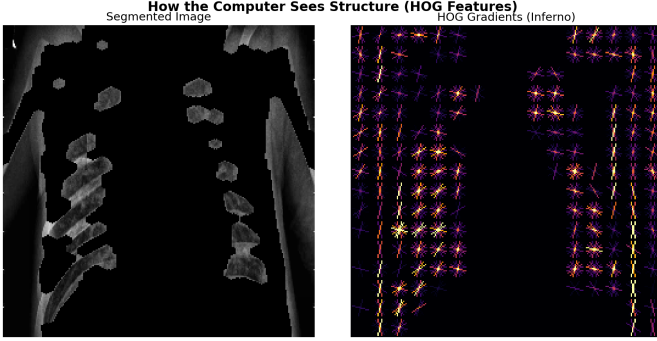


Fig. 3. Visualization of the HOG feature map extracting structural edges from the lung cavity.

IV. RESULTS AND ANALYSIS

A. Approach 1: Initial Results

The evaluation of the Traditional ML pipeline demonstrated limitations when handling clinical data. The reliance on PCA dimension reduction and SMOTE synthetics resulted in models that struggled to maintain the critical medical balance between sensitivity and specificity, validating the need for the refined Approach 2. Table I summarizes the performance metrics of the initial models evaluated under this approach.

TABLE I
EVALUATION RESULTS FOR APPROACH 1 (TRADITIONAL ML)

Model	Balanced Acc.	F1 Macro	AUC-ROC
KNN (k=1)	0.8192	0.7938	0.874
LogReg	0.6966	0.7057	0.939
SVM (RBF, C=10)	0.6855	0.6915	0.929
DecisionTree	0.6427	0.6439	0.643
RandomForest	0.6085	0.5881	0.898

B. Dimensionality Reduction Analysis: PCA vs. Raw Features

During the optimization of the final pipeline, we conducted a critical ablation study regarding dimensionality reduction. Initially, we applied Principal Component Analysis (PCA) in an attempt to compress the massive $\sim 8,000$ -dimensional HOG feature space down to 50 principal components.

However, this aggressive transformation retained only 35% of the original feature variance. Because medical imaging requires highly nuanced spatial and textural details to identify opacities, this loss of variance severely handicapped the models. Early baseline tests with PCA saw accuracy plummet

to approximately 70%. Even upon strict test set evaluation with a Balanced Random Forest, the PCA-compressed model achieved only an 84% accuracy (Macro F1: 0.82) and struggled significantly with the Normal class (Recall: 69%).

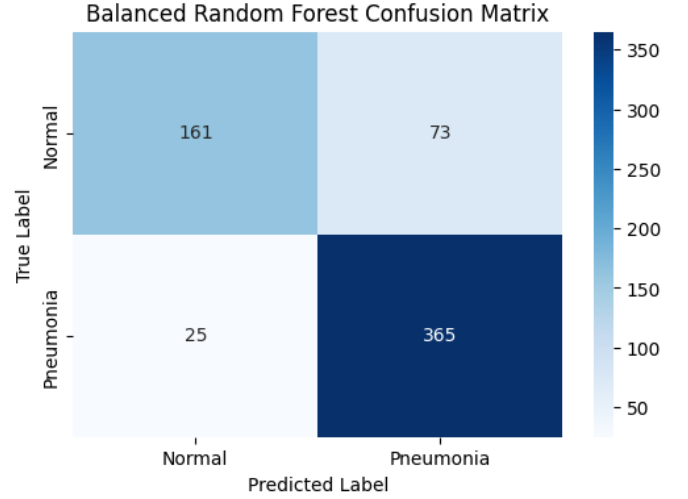


Fig. 4. Confusion Matrix for the Balanced Random Forest using PCA-reduced features. Accuracy: 84%. Note the higher rate of misclassifications.

To rectify this data loss, we abandoned PCA entirely and trained the models on the complete, raw feature vector (8,140 features). Preserving the raw features immediately boosted the overall accuracy to 89% (Macro F1: 0.87) and elevated the critical Pneumonia recall to 96%, proving that classical models in this domain benefit more from high-dimensional raw descriptors than compressed feature spaces.

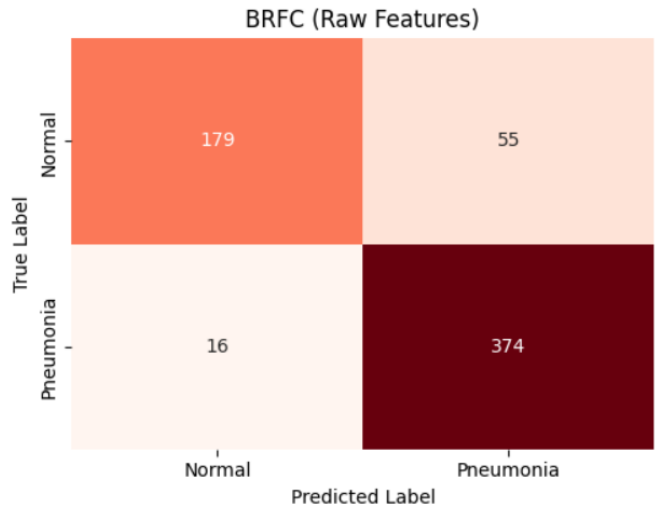


Fig. 5. Confusion Matrix for the Balanced Random Forest using Raw Features. Accuracy: 89%. Preserving raw features significantly improved clinical recall.

C. Approach 2: Final Results

Using the raw feature approach, the final models were trained using Stratified 5-Fold Cross-Validation, achieving a robust mean training accuracy of 92.79%.

The Balanced Random Forest proved to be the most clinically viable model, achieving the highest overall Macro F1-Score (0.8758).

- **Sensitivity (Pneumonia Recall): 95.90%**
- **Specificity (Normal Recall): 76.92%**
- **Overall Accuracy: 88.78%**

This model successfully identifies nearly 96% of all pneumonia cases, satisfying the primary medical objective of minimizing dangerous False Negatives.

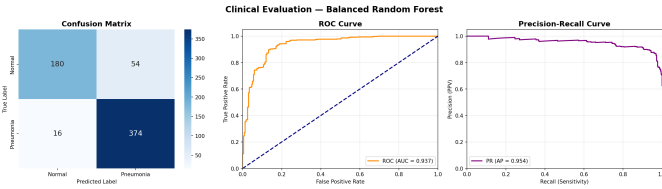


Fig. 6. Clinical evaluation metrics and confusion matrix for the Balanced Random Forest (Approach 2).

While the SVM ($C = 0.01$, RBF kernel) achieved a slightly higher specificity (86.32%), its sensitivity drop to 87.69% makes it less ideal for a strict medical screening scenario where catching the disease is the top priority.

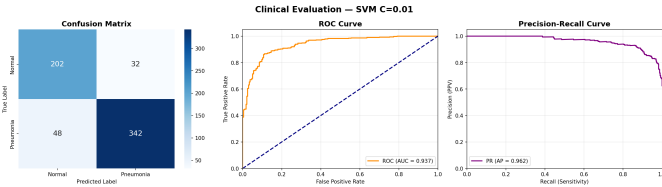


Fig. 7. Clinical evaluation metrics and confusion matrix for the Support Vector Machine (Approach 2).

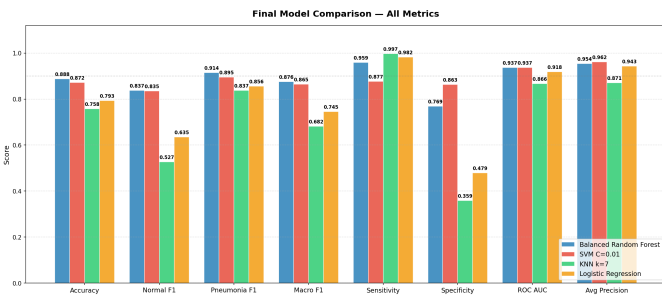


Fig. 8. A visual comparison of model performance under Approach 2, highlighting the superiority of the Balanced Random Forest.

V. CHALLENGES AND FUTURE WORK

The development of this pipeline presented several technical and dataset-level challenges. A primary initial hurdle was the severe class imbalance, heavily skewed towards Pneumonia cases.

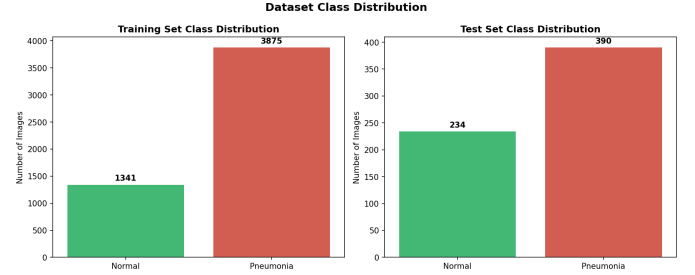


Fig. 9. Class distribution highlighting the severe imbalance between Normal and Pneumonia cases in the dataset.

A. Limitations of Approach 1

The Traditional ML pipeline faced significant limitations in a clinical context. The application of PCA for dimensionality reduction led to a critical loss of variance, discarding nuanced features necessary for distinguishing subtle pneumonia opacities. Furthermore, attempting to solve the class imbalance using SMOTE introduced synthetic noise into the dataset. These factors compounded to produce models with an unacceptable trade-off between sensitivity and specificity, rendering them clinically unviable.

B. Limitations of Approach 2

While the Clinical Imaging pipeline (Approach 2) successfully mitigated these issues through higher resolution, targeted CLAHE masking, and native algorithmic balancing (utilizing the `imblearn` library), it presents its own limitations. The reliance on handcrafted features (HOG and GLCM), although highly interpretable, requires rigid preprocessing. Additionally, the Otsu thresholding used for lung masking can occasionally struggle with severe pneumonia cases; if the lung is entirely consolidated (completely filled with fluid), it appears radiopaque (white on the X-ray) and may be incorrectly masked out as background tissue. Finally, extracting dense GLCM features is computationally heavier than the basic statistical methods used in the first approach.

C. Future Work

For future work, we plan to extend this robust backend pipeline into a fully functional, user-friendly clinical web application. Deploying the model via a Flask or FastAPI backend with a modern frontend will allow healthcare professionals to seamlessly upload X-rays and receive real-time, interpretable predictive insights.

VI. CODEBASE AND REPRODUCIBILITY

The final system is fully implemented in Python. The repository is modularized into distinct scripts for data downloading, preprocessing, feature extraction, model training, and evaluation. The environment dependencies and trained model artifacts (including the fitted Standard Scaler) are documented and version-controlled to ensure complete reproducibility of these results.